

Inventory & Order Management SaaS

Database ERD — Saari Tables, Columns & TenantId Strategy

TenantId Strategy — Ye Rule Sab Se Pehle Samjho

Multi-tenant SaaS mein sabse zaroori cheez ye hai ke ek company ka data doosri company ko kabhi dikhna nahi chahiye. Isko enforce karne ka tareeqa:

- **Organizations table** hi asal "Tenant" hai. Har business jo signup karega, usko ek **OrganizationId (TenantId)** milega.
- **Har business-data table** (Products, Customers, Orders, Inventory, waghera) mein ek column **TenantId (GUID, Foreign Key → Organizations.OrganizationId)** hoga.
- Jab bhi koi query chale (SELECT/UPDATE/DELETE), **WHERE TenantId = @CurrentTenantId** hamesha lagana hai — is se dusri company ka data kabhi mix nahi hoga.
- Best practice: EF Core mein **Global Query Filter** laga do TenantId pe, taake developer manually har jagah WHERE na likhe aur bhool na jaye.
- Sirf **system-level / global tables** (jo kisi ek tenant ki nahi, poore platform ki hain) mein TenantId **nahi** hoga — jaise Organizations khud, SubscriptionPlans, aur Permissions master list.
- Har table jismein TenantId column hai, is document mein us table ke andar **peeli row** mein highlight kiya gaya hai.

Neeche har table ka pura column structure diya hai. Jahan TenantId row peeli hai, wahan tenant isolation lagayi gayi hai.

Group 1 — Multi-Tenancy & Subscription (Global / No TenantId)

Group 2 — Authentication, Users & Roles

Group 3 — Product Catalog

Group 4 — Warehouse & Inventory

Group 5 — Customers & Suppliers

Group 6 — Sales Orders

Group 7 — Purchase Orders

Group 8 — Invoices & Payments

Group 9 — System / Audit

1. Organizations (this IS the Tenant)

Har business jo signup karti hai, uska ek record yahan banega. OrganizationId hi TenantId hai jo baaki sab tables mein use hoga.

Field	Type	Notes
OrganizationId	GUID (PK)	Primary Key — yehi TenantId hai jo aage sab jagah use hoga
Name	VARCHAR(150)	Business/company ka naam
LogoUrl	VARCHAR(255)	Company logo ka path/URL
BusinessAddress	VARCHAR(255)	Address
TaxNumber	VARCHAR(50)	NTN / Tax ID
Currency	VARCHAR(10)	PKR, USD, waghera
TimeZone	VARCHAR(50)	Business ka time zone
FiscalYearStart	DATE	Fiscal year kab shuru hota hai
DateFormat	VARCHAR(20)	dd-MM-yyyy waghera
NumberFormat	VARCHAR(20)	1,000.00 waghera
Status	VARCHAR(20)	Active / Suspended / Inactive
SubscriptionPlanId	GUID (FK)	→ SubscriptionPlans.SubscriptionPlanId
CreatedAt	DATETIME	Record creation time
UpdatedAt	DATETIME	Last update time

2. SubscriptionPlans

Platform ke saare plans (Starter, Business, Enterprise). Global table hai, kisi ek tenant ki nahi.

Field	Type	Notes
SubscriptionPlanId	GUID (PK)	Primary Key
Name	VARCHAR(50)	Starter / Business / Enterprise
MaxUsers	INT	Kitne users allowed hain
MaxWarehouses	INT	Kitne warehouses allowed hain
MaxProducts	INT	Kitne products allowed hain
Price	DECIMAL(18,2)	Monthly/Yearly price
Features	TEXT	JSON list of extra features (API access, advanced reports, etc.)

3. Users

Har organization ke users. TenantId isolation yahan se shuru hoti hai.

Field	Type	Notes
UserId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
FullName	VARCHAR(100)	User ka naam
Email	VARCHAR(150)	Login email (unique per tenant)
PasswordHash	VARCHAR(255)	Hashed password (BCrypt/Identity)
PhoneNumber	VARCHAR(20)	Contact number
EmailVerified	BOOLEAN	Email verify hua ya nahi
RefreshToken	VARCHAR(255)	JWT refresh token
RefreshTokenExpiry	DATETIME	Refresh token expiry
Status	VARCHAR(20)	Active / Inactive / Locked
CreatedAt	DATETIME	Record creation time

4. Roles

Standard roles: Super Admin, Company Admin, Manager, Salesperson, Warehouse Staff, Accountant.

Field	Type	Notes
RoleId	GUID (PK)	Primary Key
TenantId	GUID (FK, nullable)	Null = system-wide role, filled = tenant custom role
Name	VARCHAR(50)	Role ka naam
Description	VARCHAR(150)	Role ka kaam kya hai

5. Permissions

Master list of all permissions in the system (global, no TenantId).

Field	Type	Notes
PermissionId	GUID (PK)	Primary Key
Name	VARCHAR(80)	e.g. Products.Create, Orders.Cancel, Inventory.Adjust
Module	VARCHAR(50)	Products / Orders / Inventory / Reports / Users

6. RolePermissions

Kaunse role ko kaunsi permission mili hai (many-to-many link).

Field	Type	Notes
RolePermissionId	GUID (PK)	Primary Key
RoleId	GUID (FK)	→ Roles.RoleId
PermissionId	GUID (FK)	→ Permissions.PermissionId

7. UserRoles

Kis user ko kaunsa role mila hai (many-to-many link).

Field	Type	Notes
UserRoleId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
UserId	GUID (FK)	→ Users.UserId
RoleId	GUID (FK)	→ Roles.RoleId

8. Categories

Hierarchical categories (parent-child, e.g. Electronics → Mobile Phones).

Field	Type	Notes
CategoryId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
Name	VARCHAR(100)	Category ka naam
ParentCategoryId	GUID (FK, nullable)	→ Categories.CategoryId (self-reference)
Status	VARCHAR(20)	Active / Inactive

9. Brands

Product brands.

Field	Type	Notes
BrandId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
Name	VARCHAR(100)	Brand ka naam
Status	VARCHAR(20)	Active / Inactive

10. Products

Main product table.

Field	Type	Notes
ProductId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
Name	VARCHAR(150)	Product ka naam
SKU	VARCHAR(50)	Unique stock keeping unit
Barcode	VARCHAR(50)	Barcode number
Description	TEXT	Product detail
CategoryId	GUID (FK)	→ Categories.CategoryId
BrandId	GUID (FK)	→ Brands.BrandId
UnitOfMeasure	VARCHAR(20)	Piece / Kg / Box waghera
CostPrice	DECIMAL(18,2)	Purchase/cost price
SellingPrice	DECIMAL(18,2)	Default selling price
TaxId	GUID (FK, nullable)	→ Taxes.TaxId
MinimumStock	INT	Reorder ka minimum level
MaximumStock	INT	Maximum stock level
ReorderLevel	INT	Kis level pe dobara order karna hai
ImageUrl	VARCHAR(255)	Product image
Weight	DECIMAL(10,2)	Shipping weight
Dimensions	VARCHAR(50)	L x W x H
Status	VARCHAR(20)	Active / Inactive / Discontinued
CreatedAt	DATETIME	Record creation time

11. ProductVariants

Same product ke different color/size combinations (e.g. T-Shirt - Black - M).

Field	Type	Notes
VariantId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
ProductId	GUID (FK)	→ Products.ProductId
SKU	VARCHAR(50)	Variant-specific SKU
Barcode	VARCHAR(50)	Variant-specific barcode
Color	VARCHAR(30)	e.g. Black, White, Blue
Size	VARCHAR(20)	e.g. S, M, L, XL
Price	DECIMAL(18,2)	Variant price
Cost	DECIMAL(18,2)	Variant cost
ImageUrl	VARCHAR(255)	Variant image

12. Taxes

Configurable tax types (GST, VAT, Sales Tax, Custom).

Field	Type	Notes
TaxId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
Name	VARCHAR(50)	GST / VAT / Sales Tax
Percentage	DECIMAL(5,2)	e.g. 17.00
AppliesTo	VARCHAR(20)	Product / Customer / Order level

13. Warehouses

Business ke ek ya zyada warehouses/branches.

Field	Type	Notes
WarehouseId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
Name	VARCHAR(100)	e.g. Warehouse A - Islamabad
Code	VARCHAR(20)	Short code
Address	VARCHAR(255)	Warehouse address
ContactNumber	VARCHAR(20)	Contact number
ManagerUserId	GUID (FK, nullable)	→ Users.UserId
Status	VARCHAR(20)	Active / Inactive

14. Inventory

Har product/variant ka har warehouse mein current stock.

Field	Type	Notes
InventoryId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
ProductId	GUID (FK)	→ Products.ProductId
VariantId	GUID (FK, nullable)	→ ProductVariants.VariantId
WarehouseId	GUID (FK)	→ Warehouses.WarehouseId
QuantityAvailable	INT	Current available stock
QuantityReserved	INT	Stock reserved for pending orders
LastUpdated	DATETIME	Last stock change time

15. StockMovements

Har stock change ka audit trail (Purchase, Sale, Transfer, Damage, waghera).

Field	Type	Notes
StockMovementId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
ProductId	GUID (FK)	→ Products.ProductId
VariantId	GUID (FK, nullable)	→ ProductVariants.VariantId
WarehouseId	GUID (FK)	→ Warehouses.WarehouseId
MovementType	VARCHAR(30)	Purchase / Sale / Sales Return / Purchase Return / Transfer / Adjustment / Damage / Opening Stock
QuantityChange	INT	+ve ya -ve quantity
ReferenceType	VARCHAR(30)	SalesOrder / PurchaseOrder / Transfer / Adjustment
ReferenceId	GUID	Us record ki ID jis se movement bana
CreatedByUserId	GUID (FK)	→ Users.UserId
CreatedAt	DATETIME	Movement ka time

16. StockAdjustments

Manual stock correction (physical count vs system count).

Field	Type	Notes
StockAdjustmentId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
ProductId	GUID (FK)	→ Products.ProductId
WarehouseId	GUID (FK)	→ Warehouses.WarehouseId
PreviousQuantity	INT	System ka expected quantity
NewQuantity	INT	Physical count ke baad quantity
Difference	INT	NewQuantity - PreviousQuantity
Reason	VARCHAR(150)	Damaged / Lost / Miscount waghera
AdjustedByUserId	GUID (FK)	→ Users.UserId
CreatedAt	DATETIME	Adjustment ka time

17. StockTransfers

Ek warehouse se doosre warehouse mein stock bhejna (header).

Field	Type	Notes
StockTransferId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
FromWarehouseId	GUID (FK)	→ Warehouses.WarehouseId
ToWarehouseId	GUID (FK)	→ Warehouses.WarehouseId
Status	VARCHAR(20)	Draft / Requested / Approved / In Transit / Received
RequestedByUserId	GUID (FK)	→ Users.UserId
ApprovedByUserId	GUID (FK, nullable)	→ Users.UserId
CreatedAt	DATETIME	Record creation time

18. StockTransferItems

Transfer mein kaunse products, kitni quantity mein (line items).

Field	Type	Notes
StockTransferItemId	GUID (PK)	Primary Key
StockTransferId	GUID (FK)	→ StockTransfers.StockTransferId
ProductId	GUID (FK)	→ Products.ProductId
VariantId	GUID (FK, nullable)	→ ProductVariants.VariantId
Quantity	INT	Transfer ki quantity

19. Customers

Business ke customers.

Field	Type	Notes
CustomerId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
CustomerCode	VARCHAR(30)	Unique customer code
Name	VARCHAR(100)	Customer ka naam
CompanyName	VARCHAR(100)	Agar business customer hai
Email	VARCHAR(150)	Email
Phone	VARCHAR(20)	Phone number
Address	VARCHAR(255)	Address
City	VARCHAR(50)	City
TaxNumber	VARCHAR(50)	Tax number (agar business hai)
CreditLimit	DECIMAL(18,2)	Maximum udhaar limit
PaymentTerms	VARCHAR(50)	e.g. Net 30, Cash on Delivery
Status	VARCHAR(20)	Active / Inactive / Blocked

20. Suppliers

Business ke suppliers/vendors.

Field	Type	Notes
SupplierId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
SupplierCode	VARCHAR(30)	Unique supplier code
Name	VARCHAR(100)	Supplier ka naam
ContactPerson	VARCHAR(100)	Contact person ka naam
Email	VARCHAR(150)	Email
Phone	VARCHAR(20)	Phone number
Address	VARCHAR(255)	Address
TaxNumber	VARCHAR(50)	Tax number
PaymentTerms	VARCHAR(50)	Payment terms
Status	VARCHAR(20)	Active / Inactive

21. SalesOrders

Sales order ka header (customer, totals, status).

Field	Type	Notes
SalesOrderId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
OrderNumber	VARCHAR(30)	e.g. ORD-10025
CustomerId	GUID (FK)	→ Customers.CustomerId
WarehouseId	GUID (FK)	→ Warehouses.WarehouseId
OrderDate	DATETIME	Order kab bana
Status	VARCHAR(20)	Draft / Confirmed / Processing / Packed / Shipped / Delivered / Cancelled / Returned
Subtotal	DECIMAL(18,2)	Products ka total (tax/discount se pehle)
DiscountAmount	DECIMAL(18,2)	Total discount
TaxAmount	DECIMAL(18,2)	Total tax
ShippingAmount	DECIMAL(18,2)	Shipping charges
GrandTotal	DECIMAL(18,2)	Final amount
CreatedByUserId	GUID (FK)	→ Users.UserId
ApprovedByUserId	GUID (FK, nullable)	→ Users.UserId (agar approval workflow lagaya)
CreatedAt	DATETIME	Record creation time

22. SalesOrderItems

Sales order ki line items (kaunsa product, kitni quantity, price).

Field	Type	Notes
SalesOrderItemId	GUID (PK)	Primary Key
SalesOrderId	GUID (FK)	→ SalesOrders.SalesOrderId
ProductId	GUID (FK)	→ Products.ProductId
VariantId	GUID (FK, nullable)	→ ProductVariants.VariantId
Quantity	INT	Order ki quantity
Price	DECIMAL(18,2)	Per unit price
Discount	DECIMAL(18,2)	Line-level discount
Tax	DECIMAL(18,2)	Line-level tax
Total	DECIMAL(18,2)	Line total

23. PurchaseOrders

Supplier ko diya gaya purchase order (header).

Field	Type	Notes
PurchaseOrderId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
PONumber	VARCHAR(30)	e.g. PO-10025
SupplierId	GUID (FK)	→ Suppliers.SupplierId
WarehouseId	GUID (FK)	→ Warehouses.WarehouseId
OrderDate	DATETIME	PO create date
ExpectedDeliveryDate	DATETIME	Expected delivery
Status	VARCHAR(20)	Draft / Sent / Partially Received / Received / Cancelled
Subtotal	DECIMAL(18,2)	Products ka total
DiscountAmount	DECIMAL(18,2)	Total discount
TaxAmount	DECIMAL(18,2)	Total tax
GrandTotal	DECIMAL(18,2)	Final amount
Notes	TEXT	Extra notes
CreatedByUserId	GUID (FK)	→ Users.UserId
CreatedAt	DATETIME	Record creation time

24. PurchaseOrderItems

PO ki line items, aur actual receiving track karne ke liye.

Field	Type	Notes
PurchaseOrderItemId	GUID (PK)	Primary Key
PurchaseOrderId	GUID (FK)	→ PurchaseOrders.PurchaseOrderId
ProductId	GUID (FK)	→ Products.ProductId
VariantId	GUID (FK, nullable)	→ ProductVariants.VariantId
QuantityOrdered	INT	Kitni quantity order ki gayi
QuantityReceived	INT	Kitni actual mili (partial ho sakti hai)
QuantityDamaged	INT	Damaged quantity
QuantityRejected	INT	Rejected quantity
Price	DECIMAL(18,2)	Per unit purchase price
Tax	DECIMAL(18,2)	Line-level tax
Discount	DECIMAL(18,2)	Line-level discount

25. Invoices

Sales order se generate hone wala invoice.

Field	Type	Notes
InvoiceId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
InvoiceNumber	VARCHAR(30)	e.g. INV-10025
SalesOrderId	GUID (FK, nullable)	→ SalesOrders.SalesOrderId
CustomerId	GUID (FK)	→ Customers.CustomerId
Subtotal	DECIMAL(18,2)	Products ka total
DiscountAmount	DECIMAL(18,2)	Total discount
TaxAmount	DECIMAL(18,2)	Total tax
ShippingAmount	DECIMAL(18,2)	Shipping charges
GrandTotal	DECIMAL(18,2)	Final amount
PaidAmount	DECIMAL(18,2)	Ab tak kitna mila
RemainingAmount	DECIMAL(18,2)	Baqi amount
Status	VARCHAR(20)	Unpaid / Partially Paid / Paid / Overdue
DueDate	DATETIME	Payment kab tak due hai
CreatedAt	DATETIME	Record creation time

26. Payments

Customer se mila payment, ya supplier ko diya payment.

Field	Type	Notes
PaymentId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
InvoiceId	GUID (FK, nullable)	→ Invoices.InvoiceId
CustomerId	GUID (FK, nullable)	→ Customers.CustomerId (customer payment)
SupplierId	GUID (FK, nullable)	→ Suppliers.SupplierId (supplier payment)
Amount	DECIMAL(18,2)	Payment amount
PaymentMethod	VARCHAR(20)	Cash / Bank / Card / Online / Other
PaymentDate	DATETIME	Payment ki date
ReferenceNumber	VARCHAR(50)	Bank ref / transaction id
CreatedByUserId	GUID (FK)	→ Users.UserId

27. AuditLogs

Har important action ka record — kis ne kya change kiya.

Field	Type	Notes
AuditLogId	GUID (PK)	Primary Key
TenantId	GUID (FK)	→ Organizations.OrganizationId
UserId	GUID (FK)	→ Users.UserId (kis ne action kiya)
Action	VARCHAR(50)	Created / Updated / Deleted / Approved waghera
EntityName	VARCHAR(50)	e.g. Product, SalesOrder
EntityId	GUID	Us record ki ID jis pe action hua
OldValue	TEXT	Pehle ki value (JSON)
NewValue	TEXT	Nayi value (JSON)
IPAddress	VARCHAR(45)	User ka IP address
CreatedAt	DATETIME	Action ka time

Phase 2 Tables (Abhi Mat Banao — Baad Mein Add Hongi)

Ye tables MVP ke baad add hongii jab core system chal raha ho:

- **SalesReturns / SalesReturnItems** — customer returns track karne ke liye
- **PurchaseReturns / PurchaseReturnItems** — supplier ko wapis kiye gaye items
- **CustomerPricing / QuantityPricing** — customer-specific aur quantity-based pricing
- **Discounts** — promotional/order-level discount rules
- **Notifications** — in-app/email/WhatsApp alerts (low stock, order updates)

Ye sab bhi TenantId column ke sath hi banengi — jab banao to yehi pattern follow karna.